CHRISTIAN BLEVENS

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Full-stack engineer shipping ML-powered products end-to-end • Python • PyTorch • FastAPI • React • Rust

SUMMARY

- **Python, PyTorch, FastAPI, React, TypeScript, Rust, C++, C#**: full-stack delivery across ML, web, and real-time systems.
- **Deep learning & generative AI**: VAEs, transformers, diffusion, embeddings, LLMs, RAG, prompt engineering, fine-tuning, model evaluation; trained and deployed in production.
- **Cloud, backend & data**: AWS, Modal, Docker, GPU inference, Weights & Biases, CI/CD, FastAPI microservices, REST, Postgres, Redis, DuckDB, SQLite, Qdrant, GraphQL.
- **Digital signal & image processing**: Fourier analysis, filtering, OpenCV, monocular 3D reconstruction, sensor fusion, radar–camera coordinate transforms, real-time audio DSP.
- **GPU, graphics & real-time**: CUDA via PyTorch, wgpu compute shaders, Three.js, Bevy, Unity DOTS / ECS.

EDUCATION

University of Washington — Seattle, WA *Graduated June 2023*

B.S. Electrical Engineering, Digital Signal & Image Processing concentration

- **Digital signal & image processing**: Fourier analysis, FFT, filtering, sampling theory (EE 442, EE 440).
- **Machine learning for signal processing**, random signals & communications, probability & statistics, modern wireless communications, computer networks.
- **Data structures & algorithms** (CSE 373), **linear algebra & multivariable calculus** (MATH 224): applied directly to ML training, 3D rendering, and ECS systems design.
- **Digital design & FPGA fundamentals**: Verilog, hardware game capstone (EE 271, EE 371).
- **MATLAB & Simulink**: coursework in signal and system modeling and analysis.

EXPERIENCE

Machine Learning Engineer — AudeAI, Seattle, WA *Sept 2025 – Present*

- **Deep learning architecture**: designed and trained a novel disentangled VAE compressing text embeddings into a 10-dim interpretable latent space, each dim mapping to a distinct communication trait.
- **PyTorch & Weights & Biases**: built the training pipeline with per-dimension KL targets and mixed-precision GPU training; reached 0.65 cosine similarity at maximum compression with near-full disentanglement.
- **Model deployment**: shipped the VAE as a Modal-hosted GPU inference service behind a FastAPI endpoint, integrated into the team's React-based evaluator with versioned releases and rollback support.
- **Stakeholder enablement**: exposed learned latent dimensions to non-ML reviewers, lifting evaluator accuracy, consistency, evidence retrieval, and advice quality; informed prompt-strategy iteration from non-ML reviewer feedback.

Full-Stack Engineer — Voluhaus, Seattle, WA *Jan 2026 – Present*

- **Full-stack architecture**: shipped a B2B SaaS solo: Three.js front-end, FastAPI backend, DuckDB, Modal GPU inference, Cloudflare R2, Docker, nginx reverse proxy with SSL on Hetzner.
- **Computer vision & 3D reconstruction**: integrated MoGe-2 monocular depth; built surface detection and AI-driven furniture placement, calibrating camera intrinsics and surface-normal priors for placement accuracy.
- **Customer discovery & product iteration**: ran direct outreach to home-staging operators; converted pilot conversations into structured product requirements and iterated pitch from operator feedback.

Software Engineer — Contract, Seattle, WA *June 2023 – Sept 2025*

- **AWS deployment**: hosted depth-estimation and line-detection AI models powering a Christmas-light visualization web app integrated with a React + Canvas front-end; tuned batching and warm-start latency to manage cloud inference cost.
- **Computer vision pipelines**: OpenCV line detection, monocular depth, and AI image generation chained into a user-facing real-time render pipeline, latency-optimized for interactive web use.
- **Unity / C# development**: delivered contract Unity work across mobile and 3D applications, including data-oriented design, deterministic multiplayer state sync, and ECS cache-locality optimization.
- **End-to-end product delivery**: built 30+ shipped projects spanning ML, graphics, full-stack web, mobile, and infrastructure — the body of work that led directly to AudeAI.

Capstone Engineer — Echodyne (UW-sponsored), Seattle, WA *Jan 2023 – June 2023*

- **Sensor fusion & coordinate transforms**: implemented ENU radar-to-camera math driving pan/tilt/zoom servo commands for an automated radar-cued tracking camera, expanding the customer's ML training image database via radar-driven automated labeling and eliminating manual annotation cost.
- **Python + Unity simulation**: built validation visualizers and an ENU-frame test harness that caught coordinate-frame errors before hardware deployment; owned subsystem network communications.

PROJECTS (full portfolio at christianblevens.me)

- **Stock Sentiment Predictor**: Pre-norm transformer, attention-weighted multi-source sentiment, Qdrant vector store, FastAPI service, GPU AMP, Docker; per-horizon delta and confidence heads at 1d/1w/1m.
- **Arcane Physiology + Roughcast**: Multi-component Lattice Boltzmann GPU fluid sim via wgpu compute shaders; five immiscible fluids with Shan-Chen inter-component repulsion; SDF/voxel destructible world; Bevy ECS.
- **ASVAB AI Tutor SaaS**: Multi-LLM provider routing (Claude, Gemini), Stripe webhooks + credit metering, SSE streaming, JWT + Google OAuth, FastAPI, Jinja, Docker; server-pinned assessment model emitting per-concept JSON scores.
- **Live Audio Censoring**: Streaming Whisper word-level timestamps, Web Audio gain automation, end-to-end delay auto-calibration, GPU/CPU Docker; sample-accurate ducking via explicit state machines.

SKILLS

TensorFlow • scikit-learn • NumPy • Pandas • SciPy • Hugging Face • sentence-transformers • SHAP • diffusion models • LangChain • agentic frameworks • ONNX • quantization • Java • Kubernetes • Azure • Firebase • Stripe • FFmpeg • WebGL • WebSockets • SSE • Capacitor • Jinja • Rapier3D • SDF / voxel meshing • behavior trees • state machines • genetic algorithms • spatial partitioning • FABRIK / IK • Git • Linux